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# RYPE: A Human-in-the-Loop Expert-Augmented Surrogate Framework for Interpretable Cargo Disruption Risk Prediction

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***Abstract***

*Supply chain disruption risk prediction systems increasingly rely on proprietary black-box metrics, limiting interpretability, operational actionability, and human oversight. This study presents RYPE (Risk Yield Prediction Engine), motivated not merely by technical substitution but by the fundamental goal of returning structured human expertise to the prediction loop through component-level user override.*

*The dataset comprises approximately 650,000 records across 66 columns from four integrated data sources, with modelling performed on 32,065 high-risk observations. Initial models incorporating the proprietary score produced an artificial F1 = 1.00; regression diagnostics confirmed black-box dependency ($R^2 < 0$), statistically establishing data leakage. Following leakage elimination, the XGBoost model achieved weighted F1 = 0.6393 with High Risk class F1 = 0.85. The four dominant predictors were Weather Condition (14.85%), Traffic Density (11.74%), Vehicle Type (11.37%), and Region Type (8.07%).*

*The RYPE Index is defined as 0.5 × Financial Health + 0.3 × Geopolitical Risk + 0.2 × Cyber Risk, with weights validated through the Analytic Hierarchy Process (AHP, CR = 0.000, perfect consistency). World Bank Governance Indicators, UN/LOCODE port mapping, and route-specific cyber multipliers (Red Sea: ×1.4; Strait of Malacca: ×1.2) are integrated into the feature space. The human-in-the-loop architecture incorporates IQR-based input validation, a component-level user override layer, and PSI-based model drift monitoring. A cost-sensitive pipeline combining SMOTE, scale_pos_weight, and an asymmetric cost matrix addresses 95%+ High Risk concentration; SHAP attribution provides per-shipment explainability.*

*The framework makes four original contributions: (i) statistical detection and elimination of black-box data leakage in supply chain AI, (ii) AHP-validated expert-augmented composite risk index, (iii) cost-sensitive classification for asymmetric risk environments, and (iv) human-in-the-loop override management with SHAP-based per-shipment interpretability. Full RYPE-augmented quantitative results will be presented at the oral session upon completion of final optimisations.*

**Keywords:** *Supply Chain Risk, Human-in-the-Loop, Surrogate Modeling, XGBoost, Interpretable AI, Data Leakage Prevention*